The machine learning Recipe

Dataset

Maximum likelihood Estimation

(MLE) is a statistical method used in deep learning and machine learning to estimate the optimal parameters of a model.

likelihood function

$$L(\theta): \prod_{i=1}^{n} P(x_i ; \theta)$$

Why MLE is used in DL

DL models must learn parameters that represent the training data.

Negative Log-Likelihood (NLL)

$NLL = -\log L(\theta)$

correct class probability $= 0.95$

$NLL = -\log(0.95) = 0.051$

correct class probability $= 0.50$

Concept	Description	Example
MLE	Estimates model parameters by maximizing the likelihood of observed data	Neural Network training
Likelihood function	Measure how likely the observed data is under a set of parameters	probability of correct predictions.
Log-likelihood	Logarithm of the likelihood function for easier optimization	summing log probabilities
NLL	Loss function obtained by negating the log-likelihood	classification loss
cross-Entropy Loss	practical implementation of NLL for classification task	cat vs. Dog image classification.

Ex 1: Coin Toss problem

Suppose a coin is tossed 3 times. We assume the probability of getting a Head is $\theta = 0.8$ and Tail is $\theta = 0.2$ the observed results are: H(Head), H(Head), T(Tail). find the likelihood of observing this data.

Q> Calculate Error

Actual output = 1
Predicted output = 0.75
error?

Error = Target - Output
= 1 - 0.75
= 0.25

Error = 0.25

Q> find Mean Squared Error (MSE)

Target outputs
1, 0, 1
Predicted outputs
0.8, 0.3, 0.9

$$MSE = \frac{1}{3}\left[(1-0.8)^2 + (0-0.3)^2 + (1-0.9)^2\right]$$

$$= \frac{1}{3}\left[0.04 + 0.09 + 0.01\right] = \frac{0.14}{3}$$

$$= 0.0467.$$

Q> find weight update

old weight = 0.50
learning Rate = 0.20
Error = 0.40
Input = 1

New weight = oldweight + η × Error × Input

$$= 0.5 + (0.2 \times 0.4)(1)$$

$$= 0.5 + 0.08$$

$$= 0.58$$

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Linear Algebra

1. Scalar

Single numerical value has only magnitude and no direction.

2. Vector

A vector is an ordered collection of numbers arranged in a single row or single column.

Types of Vectors:

1. Vector Addition: $A = [2 \ 4 \ 6]$ $B = [1 \ 3 \ 5]$

3. Mini-Batch Gradient Descent

Instead of using one sample or the entire dataset, Mini-Batch Gradient Descent (MBGD) uses a small batch of samples.

1. Probability

the measure of how likely an event is to occur.

the probability of any event lies b/w 0 & 1.

$P(E) =$

Complement Rule:-

$P(A') = 1 - P(A)$

Ex: probability of Rain = 30

Random Variable:

1) Discrete Random variable (takes countable values)

2) Continuous Random variable. (takes any value within a range)

Probability Distribution.

- Bernouli "
- Binomial "
- Normal "
- poisson "

Ex: variance = 2.67

standard Deviation = $\sqrt{2.67}$

$= \boxed{1.63}$

Mean (Average)

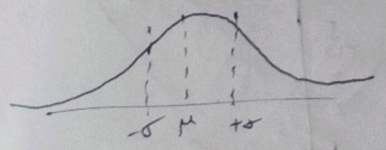
$$\boxed{\text{Mean} = \frac{\sum x}{n}}$$

Real-Time ML example

student	predicted Marks
A	70
B	75
C	80
D	85
E	90

Variance:

measures how far the data values are spread from the mean



$\text{var}(x) = E(x^2) - (E(x))^2$

Mean = $\frac{70 + 75 + 80 + 85 + 90}{5}$

$= 80.$

Variance: $\sigma^2 = N \sum (x - \mu)^2$

Standard Deviation

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1} (x_i - \mu)^2}$$

x	μ	(x-\mu)	((x-\mu)^2)
70	80	-10	100
75	80	-5	25
80	80	0	0
85	80	5	25
90	80	10	100

100 + 25 + 0 + 25 + 100 = 250

Hierarchical clustering

Unsupervised technique that create hierarchy of clusters.

A = 1, B = 2, C = 8, D = 9

d(A,B) = |1-2| = 1

d(A,C) = |1-8| = 7

d(A,D) = |1-9| = 8

d(B,C) = |2-8| = 6

d(B,D) = |2-9| = 7

d(C,D) = |8-9| = 1

d(A,B) = 1

d(C,D) = 1

merge: {A,B} & {C,D}

1. Agglomerative clustering

bottom-up approach

2. Divisive clustering

top-down

3. Dimensionality Reduction

process of reducing the no. of terms.

4. Principal Component Analysis (PCA)

dimensionality reduction technique that transforms a large no of correlated feature into a smaller no of uncorrelated variables called PCA.

Representation learning

automatically discovering and extracting meaningful features from raw data. helps a ML model make accurate predictions.

Raw data
↓
Representation learning
↓
useful feature Representation
↓
ML Model
↓
final prediction.

Types:

1. Low-Level features
2. Mid-level features.
3. High-level features.

Neural Networks Design

Computational model consisting of interconnected neurons organized into layers.

input layer = 4 neurons
Hidden layer 1 = 8 neurons
2 = 6 neurons
output layer = 2 neurons

Depth of a Neural Network

depth of a refers to no. of layers, especially the no. of computational or hidden layer in the network.

Model Capacity:

ability of neural network to learn & represent complex patterns and functions from data.

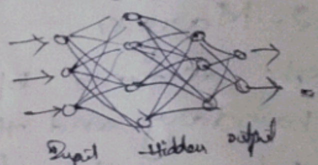
Network A 5 → 4 → 2
i/p hidden o/p

Network B 5 20 → 15 → 10 → 2
i/p hidden o/p

Network Architecture

Feed forward Neural Network (FNN)

- flow in 1 direction
- no feed back loops
- Used for classification & regression



CNN

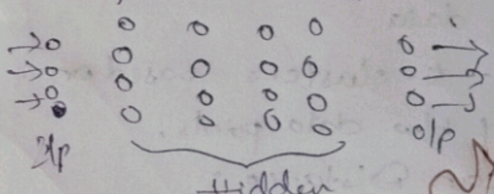
1. Input Layer
2. Convolutional layer
3. Activation layer
4. Pooling layer.

Recurrent Neural Network (RNN)

to process sequential data

Deep Neural Networks

Multiple hidden layers.



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